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UNIVERSITY OF MYSORE

(Re-accredited by NAAC with 'A' Grade)

(NIRF-2025: Ranked 20th in University Category and 71st in Overall Category)

MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri Campus, University of Mysore, Mysuru

(Approved by AICTE, New Delhi)

Mini Project

(21AIP67)

Report

On

"HealthLens-W: AI-Powered Facial Analysis for Women's Hormonal Health"

A Report submitted to

Department of Artificial Intelligence and Machine Learning in fulfillment of the Project Phase-I (21AIP67) for the award of the Bachelor of Engineering degree in Artificial Intelligence and Machine Learning

By

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Jun 2026

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CERTIFICATE

This is to certify that the **Mini Project (21AIP67)** entitled **"HealthLens-W: AI-Powered Facial Analysis for Women's Hormonal Health"** is a Bonafide work carried out by **AISHWARYA S, DEEKSHA K.N, LASYAPRIYA A.P and MANYASHREE M.J,** VI Semester students bearing Register Numbers **23SEAI003, 23SEAI17,23SEAI022 and 23SEAI024**, respectively, of the Department of **Artificial Intelligence and Machine Learning**, in partial fulfillment for the award of the degree of **Bachelor of Engineering** at the **Mysore University School of Engineering, University of Mysore, Mysuru.**

It is further certified that all corrections and suggestions indicated during the internal evaluation have been duly incorporated by the above-mentioned students.

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DECLARATION

We, **AISHWARYA S, DEEKSHA K.N, LASYAPRIYA A.P and MANYASHREE M.J,** bearing Register Numbers **23SEAI003, 23SEAI017, 23SEAI022, and 23SEAI024** respectively, students of VI Semester, **Bachelor of Engineering** in the Department of Artificial Intelligence and Machine Learning, University of Mysore, Mysuru, hereby declare that the **Mini Project (21AIP67)** entitled **"HealthLens-W: AI-Powered Facial Analysis for Women's Hormonal Health"** has been duly carried out by us under the guidance of **Ms. Keerthana N.**The Mini Project report of the above entitled work is submitted by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** by the Department of Artificial Intelligence and Ma-chine Learning, Mysore University School of Engineering, University of Mysore, Mysuru, during the academic year **2025–26**. Further, the matter embodied in this report has not been submitted previously by us or by any other person for the award of any degree, diploma, or certificate programme.

Date:23/06/2026
Place: Mysuru

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ABSTRACT

HealthLens is an innovative AI-powered facial analysis system designed to assist women in monitoring their hormonal health through non-invasive facial assessments. Hormonal imbalances often manifest through visible facial indicators such as acne, pigmentation, dark circles, skin texture changes, facial hair growth, and puffiness. By leveraging advanced Artificial Intelligence, Computer Vision, and Machine Learning techniques, HealthLens analyzes facial images to detect these subtle signs and provide personalized health insights.

The system captures and processes facial images, extracts relevant skin and facial features, and compares them with trained datasets to identify patterns associated with common hormonal conditions such as Polycystic Ovary Syndrome (PCOS), thyroid disorders, estrogen imbalance, and stress-related hormonal fluctuations. Based on the analysis, users receive health risk indicators, lifestyle recommendations, skincare suggestions, and guidance on seeking professional medical consultation when necessary.

HealthLens aims to empower women with accessible, affordable, and early health monitoring tools, enabling proactive management of hormonal well-being. By combining AI-driven analysis with user-friendly technology, the platform promotes awareness, early detection, and informed healthcare decisions while maintaining user privacy and data security. This approach has the potential to improve women's healthcare outcomes by providing continuous, convenient, and personalized hormonal health assessment.

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Contents

ABSTRACT ii

LIST OF ABBREVIATIONS vii
viii

1 INTRODUCTION 1
1.1 Origin and History of Homonal Disorder Detection 1
1.1.1 Overview of Hormonal Categories 1
1.1.2 Causes and Risk Factors 2
1.1.3 Advantages of AI-Driven Early Detection 2
1.1.4 Limitations of Traditional Diagnostic Methods 2
1.1.5 Role of Artificial Intelligence in Speech Disorder Detection 2
1.2 Background, Clinical Overview, and Technological Paradigm 2
1.3 Challenges and Limitations of AI-Based Systems 3
1.4 Hormonal Signal Illustration 4
1.5 Related Work 4
2.2 State of Art 6

2 PROBLEM STATEMENT AND SYSTEM DESIGN 7
2.1 Problem Statement 7
2.2 Core Challenges 7
2.3 Existing System 8
2.4 Proposed System 8
2.5 Functional Modules of the Proposed System 10
2.6 System Requirements 10
2.6.1 Hardware Requirements 10
2.6.2 Software Requirements 11

3 DATASET 12
3.1 Dataset Source 12
3.2 Dataset Composition 12
3.3 Audio Data Description 12
3.4 Text Data Description 13
3.5 Image Disorder Classes 13
3.6 Data Cleaning and Preparation 14

4 OBJECTIVES 19
4.1 Objectives 19
4.1.1 Text Data Preprocessing Pipeline 19
4.1.2 Hormonol Disorder Detection Model 19

4.1.3 Model Evaluation and Explainability 19
4.2 Text Data Preprocessing Pipeline 20
4.3 Model Evaluation and Validation 21

5 DATA PREPROCESSING PIPELINE **22**
5.1 Preamble 22
5.2 Introduction 22
5.3 Objective of Data Preprocessing 23
5.4 Dataset Understanding 23
5.5 Data Preprocessing Pipeline 23
5.6 Architecture of the Preprocessing Pipeline 24
5.7 Data Cleaning and Handling Missing Values 25

REFERENCES **30**

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
XGBoost	Extreme Gradient Boosting
PCOS	Polycystic Ovary Syndrome
PCOD	Polycystic Ovarian Disease
BMI	Body Mass Index
TSH	Thyroid Stimulating Hormone
T3	Triiodothyronine Hormone
T4	Thyroxine Hormone
FTI	Free Thyroxine Index
EDA	Exploratory Data Analysis
MobileNetV2	Mobile Neural Network Version 2
GPU	Graphics Processing Unit
UI	User Interface
API	Application Programming Interface
CSV	Comma-Separated Values
JPEG	Joint Photographic Experts Group
PNG	Portable Network Graphics
RGB	Red Green Blue
ReLU	Rectified Linear Unit

CHAPTER 1

INTRODUCTION

1.1 Origin and History of Speech Disorder Detection

Hormonal disorder prediction has evolved from traditional clinical diagnosis to advanced Artificial Intelligence (AI)-based healthcare systems. For many years, disorders such as PCOS, PCOD, and thyroid diseases were diagnosed through clinical examinations, blood tests, ultrasound scans, and specialist consultations. Although these methods provide accurate results, they are often time-consuming, expensive, and inaccessible to many individuals. As a result, early symptoms are frequently ignored, leading to delayed diagnosis and treatment.

The emergence of Machine Learning (ML) and Deep Learning (DL) technologies has transformed healthcare by enabling automated disease prediction from medical and lifestyle data. Modern AI systems can analyze symptoms, menstrual cycle information, health history, and medical images to identify patterns associated with hormonal disorders. Today, AI-powered health prediction systems offer scalable and efficient solutions for early detection and preventive healthcare.

1.1.1 Overview of Hormonal Disorders

Hormonal disorders occur when the body produces too much or too little of certain hormones, affecting normal physiological functions. The proposed system focuses on:

- PCOS (Polycystic Ovary Syndrome)
- PCOD (Polycystic Ovarian Disease)
- Thyroid Disorders

Common symptoms include:

- Irregular menstrual cycles
- Weight gain
- Acne and skin problems
- Hair loss or excessive hair growth
- Fatigue and mood change

1.1.2 Causes and Risk Factors

Hormonal disorders may develop due to several biological and environmental factors.

Common risk factors include:

- Genetic predisposition
- Obesity and unhealthy lifestyle habits

- Stress and inadequate sleep
- Poor dietary patterns
- Insulin resistance and metabolic disorders

1.1.3 Importance of Early Detection`

Early detection of hormonal disorders is essential for preventing long-term complications and improving women's health outcomes.

- Timely medical intervention
- Reduced risk of infertility
- Better metabolic health management
- Improved quality of life
- Increased awareness of hormonal health

1.1.4 Limitations of Traditional Diagnostic Methods

Traditional diagnostic procedures face several challenges:

- Dependence on laboratory testing and clinical examinations
- High cost of diagnosis
- Time-consuming procedures
- Limited accessibility in rural areas
- Delayed identification of symptoms

1.1.5 Role of Artificial Intelligence in Hormonal Health Prediction

Artificial Intelligence enables automated analysis of health-related information and medical images.AI systems can:

- Analyze symptom-based data using Machine Learning algorithms
- Detect acne severity from facial images using Deep Learning models
- Identify hidden patterns associated with hormonal disorders
- Provide quick and reliable predictions
- Assist users in seeking timely medical consultation

1.1.6 Background, Clinical Overview, and Technological Paradigm

Hormonal disorders have become one of the most common health concerns among women worldwide. Conditions such as PCOS, PCOD, and thyroid disorders affect reproductive health, metabolism, and overall well-being. These disorders often present gradual and overlapping symptoms, making diagnosis challenging during the early stages.

Traditional healthcare systems rely heavily on blood tests, ultrasound scans, and clinical evaluation. While effective, these methods may not always be practical for regular screening. Recent advances in Artificial Intelligence, Machine Learning, and Deep Learning have introduced new opportunities for automated health assessment

2 Challenges and Limitations of AI-Based Prediction Systems

2.1 Challenges

- Limited availability of high-quality hormonal health datasets
- Variations in acne image quality and lighting conditions
- Similarity of symptoms across different hormonal disorders
- Data imbalance among disease categories
- Privacy and security concerns in healthcare applications

2.2 Limitations

- Prediction accuracy depends on dataset quality
- AI predictions cannot replace professional medical diagnosis
- MobileNetV2 performance may vary with image quality
- Users may provide incomplete symptom information

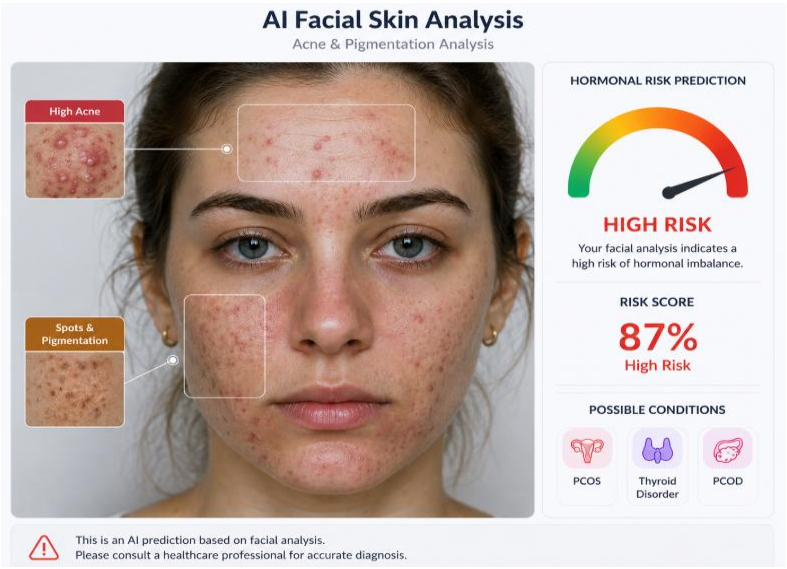


Fig :1.1 facial skin analysis

2.3 Related Work

Recent advancements in Artificial Intelligence (AI) have enabled the development of automated systems for predicting hormonal disorders such as PCOS, PCOD, and thyroid diseases. Researchers have applied Machine Learning algorithms including Random Forest, Support Vector Machine (SVM), and XGBoost to analyze symptoms and health-related data for early disease

prediction.Deep Learning models such as CNNs and MobileNetV2 have also been used for acne detection and skin analysis, as acne is often associated with hormonal imbalances. Recent studies show that combining symptom-based prediction with image-based analysis improves overall prediction accuracy and reliability.

Despite promising results, challenges such as limited datasets, overlapping symptoms, and variations in image quality continue to affect system performance. These limitations motivate the development of the proposed Women HealthLens system, which integrates XGBoost and MobileNetV2 for comprehensive hormonal disorder prediction.

2.4 State of Art

Sl.	Technique	Year	Performance / Observation
1	Machine Learning-Based PCOS Prediction	2025	Enabled early prediction using symptom and lifestyle data.
2	Random Forest Classification	2025	Achieved good accuracy in hormonal disorder prediction.
3	XGBoost-Based Prediction Models	2025	mproved prediction accuracy and handled complex health data efficiently.
4	CNN-Based Acne Detection	2025	Successfully identified acne severity from facial images.
5	MobileNetV2 for Skin Analysis	2025	Provided efficient image classification with reduced computational cost.
6	Multimodal Healthcare Systems	2025	Combined symptom and image analysis for better prediction performance.
7	Explainable AI in Healthcare	2026	Improved transparency and interpretability of AI predictions.
8	AI-Based Hormonal Disorder Screening	2026	Combined high classification accuracy with clinically interpretable predictions Enabled accessible and scalable early detection of PCOS, PCOD, and thyroid disorders.

Table 1.1: State-of-the-Art Comparison of Hormonal Disorder Prediction Techniques

Recent advancements in hormonal disorder prediction have demonstrated significant progress through Machine Learning and Deep Learning techniques. Modern systems utilize algorithms such as Random Forest, XGBoost, Convolutional Neural Networks (CNNs), and MobileNetV2 to analyze symptom-based data and facial skin conditions associated with hormonal imbalances. Multimodal approaches that combine health information with image analysis have shown improved prediction accuracy and reliability. Despite these advancements, existing systems face challenges such as limited healthcare datasets, overlapping symptoms among disorders, variations in image quality, and limited interpretability of predictions. These limitations highlight the need for an accurate, accessible, and intelligent framework that integrates symptom analysis and acne image classification for the early prediction of PCOS, PCOD, and thyroid disorders.

CHAPTER 2

PROBLEM STATEMENT AND SYSTEM DESIGN

3.1 Problem Statement

Hormonal disorders such as PCOS, PCOD, and thyroid diseases are among the most common health issues affecting women worldwide. These disorders often exhibit symptoms such as irregular menstrual cycles, acne, weight gain, fatigue, and hormonal imbalances. Since many of these symptoms overlap across different conditions, early identification becomes difficult without proper medical testing and specialist consultation.

Traditional diagnostic methods rely on blood tests, ultrasound scans, and clinical examinations. Although effective, these procedures can be expensive, time-consuming, and inaccessible to individuals in rural or underserved regions. Delayed diagnosis may result in complications such as infertility, metabolic disorders, and reduced quality of life.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have enabled automated healthcare prediction systems. However, many existing solutions focus either on symptom analysis or image analysis alone, limiting prediction accuracy and reliability.Therefore, there is a need for an intelligent and integrated prediction system that can analyze both symptom-based information and acne images to provide early insights into hormonal health conditions. The proposed Women HealthLens system utilizes XGBoost and MobileNetV2 models to predict the risk of PCOS, PCOD, and thyroid disorders, thereby supporting early awareness and preventive healthcare

.

3.1.1 Core Challenges

The major challenges involved in hormonal disorder prediction are summarized as follows:

1. **Symptom Overlap:** PCOS, PCOD, and thyroid disorders often share similar symptoms, making accurate prediction difficult.
2. **Data Quality and Availability**: Healthcare datasets may contain missing values, inconsistencies, and imbalanced disease categories.
3. **Image Variability:** Acne images may vary in lighting conditions, camera quality, and skin tone, affecting model performance.
4. **Multimodal Data Integration**: Combining symptom data and image data effectively is essential for improving prediction accuracy.

3.1.2 Existing System

Existing hormonal disorder diagnosis primarily depends on clinical examinations, blood tests, ultrasound scans, and specialist consultation. While these methods are widely accepted, they have several limitations:

• Dependence on laboratory testing and healthcare professionals.

• High diagnostic cost and time consumption.

• Limited accessibility in remote and rural regions.

• Delayed detection due to lack of awareness and regular screening.

• Most AI-based systems focus only on symptoms or image analysis individually.

• Limited integration of visual and clinical indicators for comprehensive prediction.

3.1.3 Proposed System

To overcome the limitations of conventional diagnostic approaches, this project proposes Women HealthLens, an AI-powered hormonal disorder prediction system. The framework combines Machine Learning and Deep Learning techniques to analyze both symptom-based information and acne images.The system begins with data collection, where users provide health-related information such as age, menstrual cycle details, lifestyle factors, and symptoms associated with hormonal disorders. Simultaneously, facial acne images are collected for visual analysis.

During preprocessing, structured data undergoes cleaning, encoding, and normalization, while images are resized and normalized for deep learning analysis. The processed symptom data is analyzed using the XGBoost algorithm, which predicts the likelihood of PCOS, PCOD, and thyroid disorders. The acne images are analyzed using MobileNetV2 to classify acne severity and identify visual indicators associated with hormonal imbalance.
The outputs from both modules are integrated to generate a comprehensive prediction. The final system provides risk levels and health insights that help users understand potential hormonal health conditions and seek timely medical consultation.Compared to traditional approaches, the proposed system offers faster prediction, improved accessibility, multimodal analysis, and enhanced support for early detection and preventive healthcare.

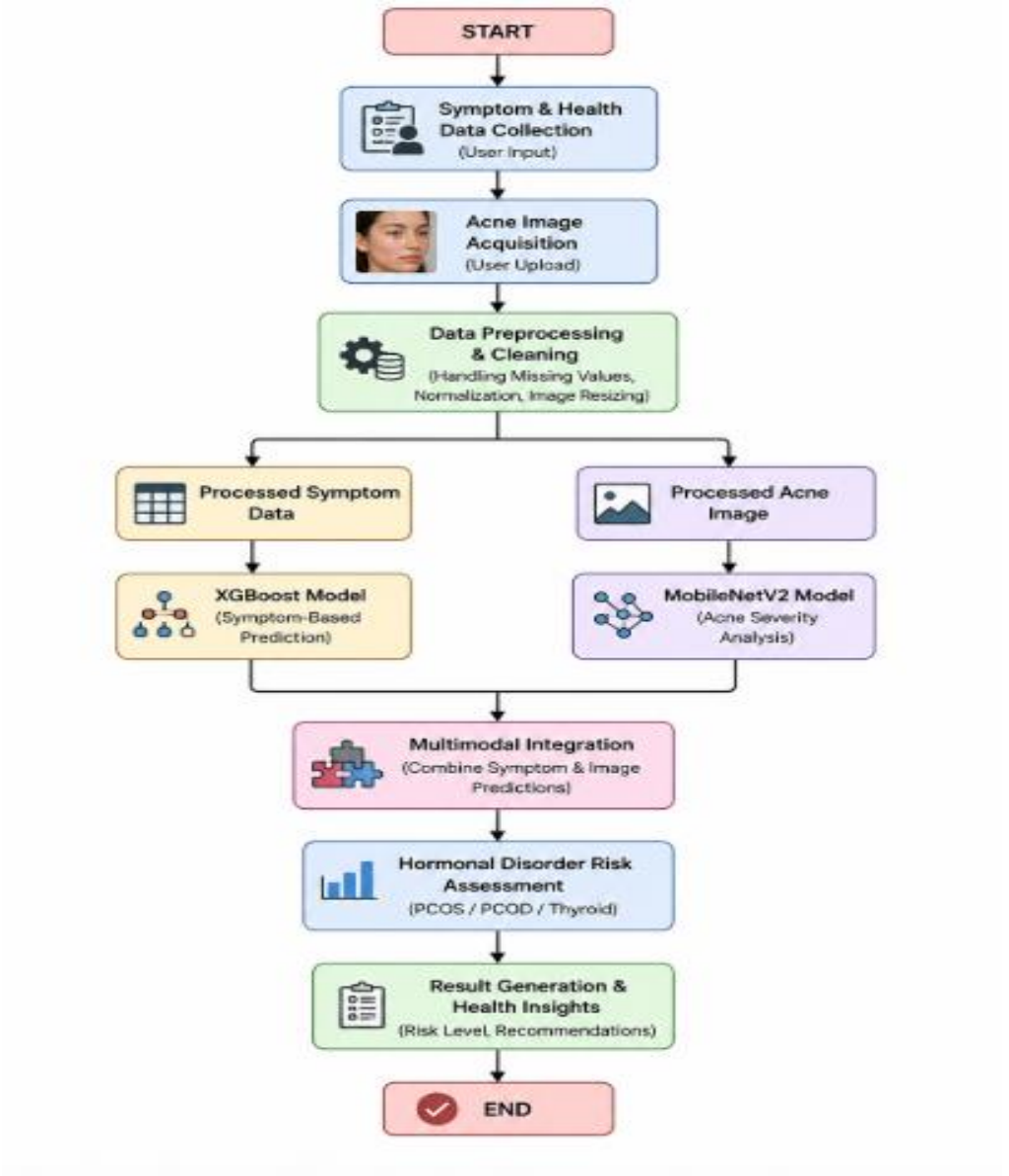


Figure 2.1: End-to-End Hormonal Disorder Detection Pipeline

The processed symptom data and acne images are analyzed through separate prediction modules. The Text-Based Prediction Module utilizes the XGBoost algorithm to analyze structured information such as symptoms, menstrual cycle details, lifestyle factors, and medical history. Simultaneously, the Image-Based Prediction Module employs MobileNetV2 to analyze acne images and identify visual indicators associated with hormonal imbalance.

The outputs generated by both modules are integrated to form the final prediction. This multimodal approach enables the system to consider both internal symptoms and external visual indicators, thereby improving prediction accuracy and reliability.

Finally, the system accepts user health information and acne images as input and generates predictions indicating the likelihood of PCOS, PCOD, and thyroid disorders. The system also provides health insights and prediction results that can assist users in seeking timely medical consultation. By integrating

machine learning and deep learning techniques, the proposed Women HealthLens system offers a more accessible, efficient, and comprehensive approach to hormonal disorder prediction compared to traditional diagnostic methods.

2.1 Functional Modules of the Proposed System

The proposed system consists of the following major functional modules:

1. Symptom and Health Data Collection
2. Acne Image Acquisition
3. Data Preprocessing and Cleaning
4. Symptom-Based Prediction using XGBoost
5. Acne Severity Analysis using MobileNetV2
6. Multimodal Prediction Integration
7. Hormonal Disorder Risk Assessment
8. Result Generation and Health Insight Display

2.2 System Requirements

2.2.1 Hardware Requirements

Component	**Specification**
Processor	Intel Core i5/i7 or equivalent
RAM	Minimum 8 GB
Storage	Minimum 20 GB free space
Camera	Smartphone Camera or Webcam
Internet	Required for dataset access and deployment
Operating System	Windows 10/11 or Linux

Table .1.2 : Hardware Requirements

2.2.2 Software Requirements

Component	Specification
Operating System	Windows / Linux / macOS
Programming Language	Python 3.8 or above
Development Environment	Google Colab / Jupyter Notebook / VS Code
Deep Learning Frameworks	TensorFlow, Keras
Machine Learning Libraries	XGBoost, Scikit-learn
Data Processing Libraries	NumPy, Pandas
Image Processing Libraries	OpenCV, Pillow
Visualization Tools	Matplotlib, Seaborn
Dataset Source	Acne Image Dataset, PCOS/PCOD/Thyroid Health Dataset

Table 1.3: Software Requirements for Women HealthLens System

CHAPTER 3

DATASET

3.1 Dataset Source

The present research utilizes a facial image dataset collected from publicly available sources and organized for hormonal health analysis. The dataset contains facial images of women categorized into different hormonal health conditions based on associated clinical and physiological indicators.

The dataset was prepared to support the development of an AI-powered facial hormonal analysis system capable of identifying potential hormonal imbalances using deep learning techniques. Each facial image is assigned an appropriate class label and undergoes preprocessing before model training.

3.2 Dataset Composition

The dataset consists of facial images belonging to three hormonal health categories:

- Low Hormonal Risk
- Moderate Hormonal Risk
- High Hormonal Risk

Each image represents a frontal facial view and is associated with a class label indicating the hormonal risk level.

This structure enables automated feature extraction and classification using deep learning models

3.3 Image Data Description

- The dataset contains high-quality facial images collected under different lighting conditions, facial expressions, and image backgrounds.
- **Image Format:** JPG (.jpg), PNG (.png)
- **Image Resolution:** Resized to 128 × 128 pixels
- **Image Content:** Frontal facial images of women
- **Preprocessing Applied:**
- Face detection and cropping

- Image resizing
- Pixel normalization
- Data augmentation
- **Feature Extraction:**
- Deep facial features extracted using MobileNetV2
- CNN-based feature representation

Attribute	**Category**	**Description**
ImageType	Facial image	Facial photography
Resolution	128*128	Input image size
Image format	JPG/PNG	Standard image format
Hormonal class	H/M/L	Target class label

Table 1.4 : Facial Image Dataset Summary

3.4 Textual Data Description

In addition to facial images, the dataset contains structured textual information associated with each subject. These attributes provide valuable contextual information that supports facial image analysis and improves the performance of the hormonal health prediction system.

The textual data includes demographic information, physiological measurements, and hormonal health indicators. These attributes are used to supplement image-based features and enhance classification accuracy.

The dataset information includes age, body mass index (BMI), hormonal risk category, and other relevant health-related attributes

Attribute	**Category**	**Description**
Age	Numeric	Age of the indivdual
BMI	Numeric	Body Mass Index
Hormonal risk	Low/Medium/High	Target class label
Image ID	Text	Unique image identifier
Subject ID	Text	Unique participant identifier
Health indicators	Categorical	Associated hormonal characteristics

Table 1.5: Text Dataset Summary

3.5 Hormonal Health Classes

The facial image dataset is categorized into three classes:

Low Hormonal Risk

Images exhibiting minimal facial indicators associated with hormonal imbalance.

Moderate Hormonal Risk

Images containing moderate facial characteristics associated with hormonal fluctuations.

High Hormonal Risk

Images showing prominent facial indicators that may suggest hormonal imbalance.

This categorization enables automated classification of hormonal health conditions.

3.6 Data Cleaning and Preparation

To ensure data quality and model readiness, the following preprocessing steps were applied **:**

Removal of Corrupted and Duplicate Images:

Invalid, damaged, and duplicate images were removed to improve dataset quality.

Face Detection and Facial Region Extraction:

Faces were detected and cropped to focus on important facial features.

Image Resizing (128 × 128 Pixels):

All images were resized to a fixed size of 128 × 128 pixels for consistency.

Pixel Value Normalization:

Pixel values were scaled to a standard range to improve model performance.

Data Augmentation (Rotation, Zooming, Flipping):

Additional image variations were generated to increase dataset diversity.

Class Label Verification:

Labels were checked to ensure each image belonged to the correct hormonal condition category

3.7.1 Image-Based Features

- **Facial Landmark Features:**
  Key facial points were identified to analyze facial structure and patterns.
- **Skin Texture Features:**
  Texture information was extracted to detect acne, pigmentation, and skin irregularities.
- **Facial Region Features:**
  Features from the forehead, cheeks, jawline, and chin were analyzed for hormonal indicators.
- **CNN Deep Features:**
  High-level image features were automatically extracted using the CNN model.

3.7.2 Text-Based Features

- **Symptom Features:**
  User-reported symptoms such as acne, irregular periods, and hair growth were utilized.
- **BMI Category Features:**
  BMI-related information was included as an important hormonal health indicator.
- **Lifestyle Features:**
  Factors such as exercise habits, sleep patterns, and dietary information were considered.

- **Encoded Text Features:**

Categorical text data were converted into numerical form for machine learning processing.

3.7 Dataset Usage in the Proposed System

The final processed dataset is organized as a collection of acoustic feature vectors and Mel-Spectrogram representations extracted from speech recordings. These features capture both spectral and temporal information present in children's speech.

The dataset serves as the primary input for the proposed Hybrid CNN-BiLSTM framework, where the CNN component learns spatial features from spectrogram representations and the BiLSTM component captures temporal dependencies present in speech sequences. This combination enables accurate identification of speech disorders from speech recordings.

3.8 Significance of the Dataset

The facial image dataset provides valuable visual information related to hormonal health conditions. The dataset supports:

- Non-invasive hormonal health assessment
- Automated facial feature analysis
- Early risk identification
- AI-assisted healthcare screening

3.9 Sample Dataset Structure

Image ID	Subject ID	Class	Resoultion
IMG001	S001	Low	128 × 128
IMG002	S002	Moderate	128 × 128
IMG003	S003	High	128 × 128
IMG004	S004	Moderate	128 × 128
IMG005	S005	Low	128 × 128

Table 1.6: Sample Dataset Structure

3.10 Class Mapping

Class Label	Description
0	Low risk
1	Moderate
2	High Risk

Table 1.7: Sample Speech Class Mapping

3.11 Facial Feature Representation

Feature	Skin texture information
Texture Features	Facial geometry patterns
Structural Features	Deep feature embeddings
CNN Features	Facial balance indicators
Symmetry Features	Overall facial representation
Global Features	Skin texture information

Table 1.8 presents the major features used in the proposed system

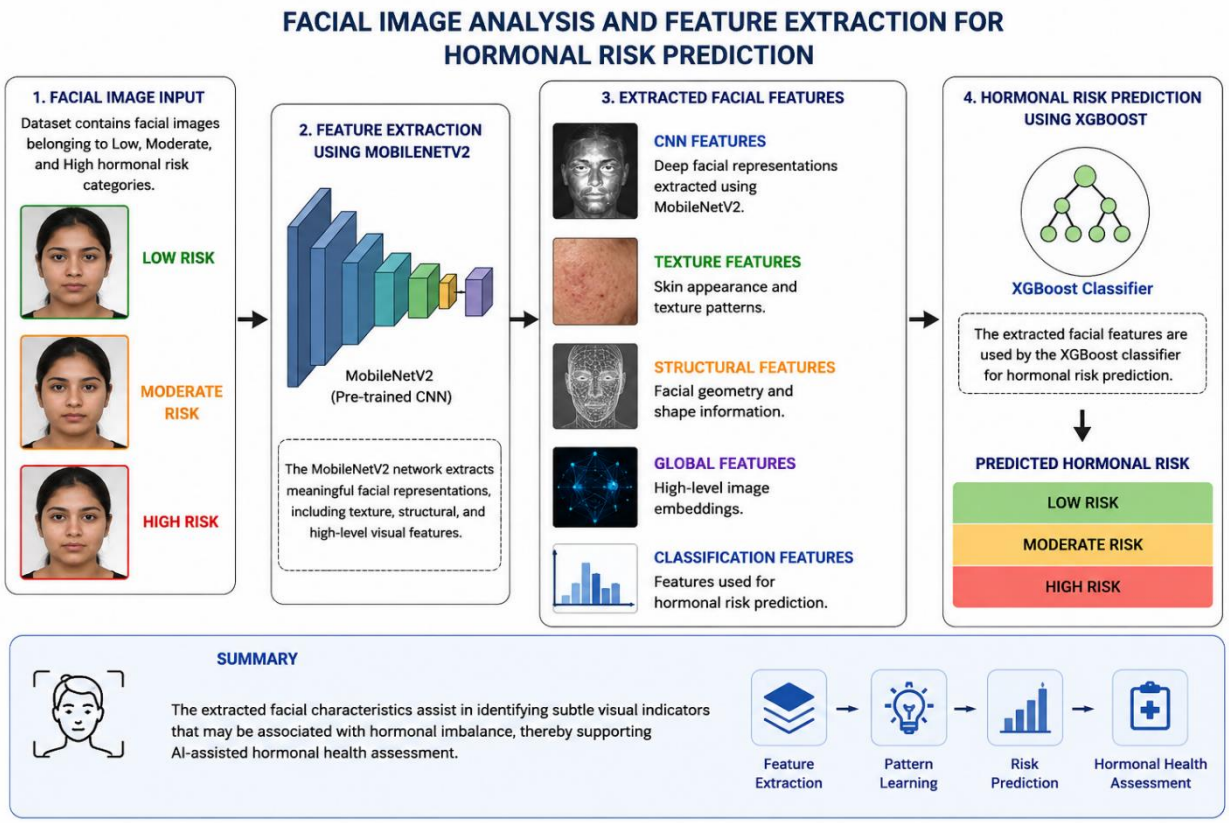
3.12 Sample Facial Image Representation

The facial image dataset contains images belonging to Low, Moderate, and High hormonal risk categories. The facial images are processed using deep learning techniques to identify patterns associated with hormonal health conditions.

The MobileNetV2 network extracts meaningful facial representations, including texture, structural, and high-level visual features. These features are subsequently used by the XGBoost classifier for hormonal risk prediction.

The extracted facial characteristics assist in identifying subtle visual indicators that may be associated with hormonal imbalance, thereby supporting AI-assisted hormonal health assessment.

- CNN Features: Deep facial representations extracted using MobileNetV2.
- Texture Features: Skin appearance and texture patterns.
- Structural Features: Facial geometry and shape information.



- Global Features: High-level image embeddings.
- Classification Features: Features used for hormonal risk prediction.

CHAPTER 4

OBJECTIVES

1.1 Objectives

Facial Image Data Collection and Preprocessing

Develop an automated pipeline for collecting and preprocessing facial images to support hormonal disorder analysis.

• Collect facial images from publicly available datasets and user-uploaded images.

• Perform image preprocessing operations such as resizing, normalization, and noise removal.

• Detect and crop facial regions using computer vision techniques.

• Enhance image quality for reliable feature extraction.

• Prepare standardized facial images for deep learning analysis.

Facial Feature Extraction and Analysis

Extract meaningful facial characteristics associated with hormonal imbalances.

• Analyze acne distribution and severity patterns.

• Detect facial pigmentation and skin discoloration.

• Identify facial hair growth patterns related to hormonal disorders.

• Extract texture, color, and skin-related features.

• Generate deep facial feature representations using Convolutional Neural Networks (CNN).

4.1.4 Model Evaluation and Explainability

Evaluate the effectiveness and reliability of the proposed system.

• Assess model performance using Accuracy, Precision, Recall, F1-Score, AUC-ROC, and Confusion Matrix.

• Compare CNN-XGBoost performance with conventional machine learning approaches.

• Apply Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME.

• Identify facial features that contribute most to hormonal disorder prediction.

• Improve transparency and trustworthiness of AI-generated predictions.

Model Evaluation and Validation Pipeline

The preprocessing pipeline converts raw facial images into processed feature representations that can be used for hormonal disorder prediction using CNN and XGBoost models**.**

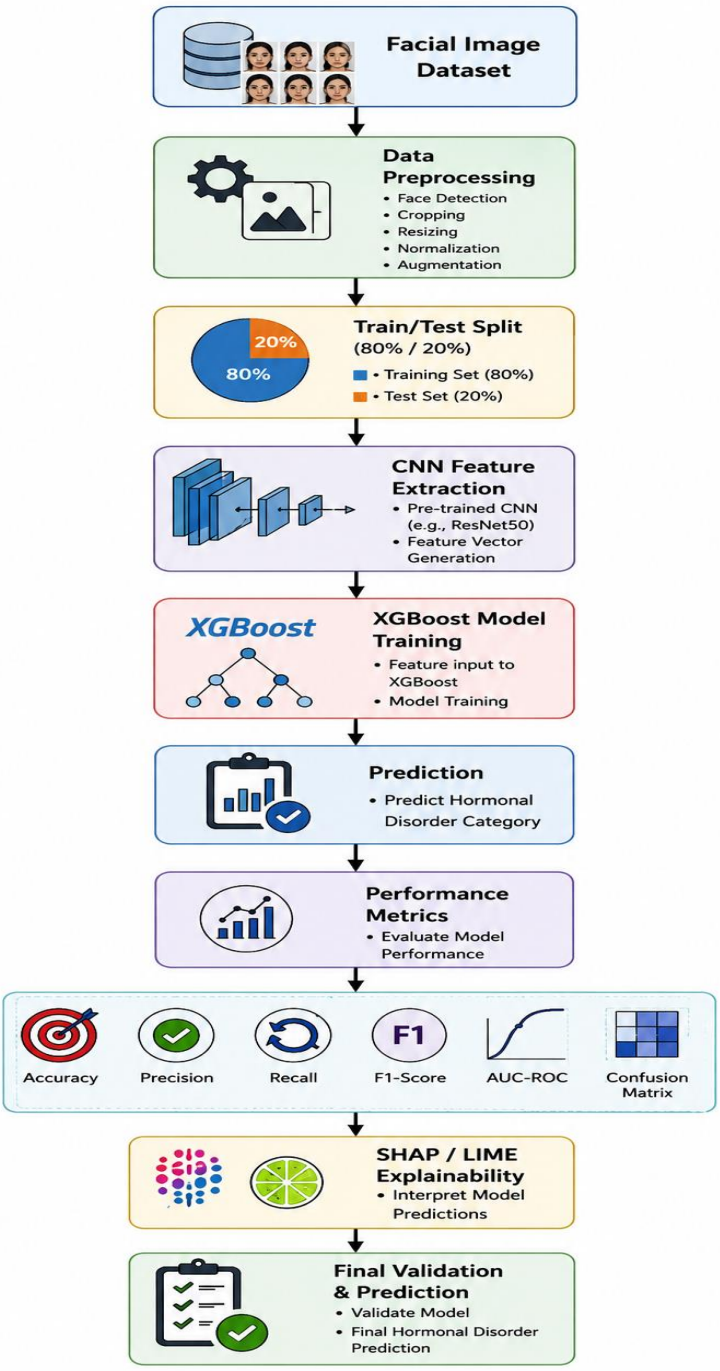


Figure 4.1: Hormonal Data Preprocessing Pipeline

CHAPTER 5

DATA PREPROCESSING PIPELINE

5.1 Model Training Configuration

The proposed hormonal health risk prediction system utilizes **MobileNetV2** as the feature extraction network. Transfer learning was employed by loading ImageNet pretrained weights, while the final classification layers were customized for the three hormonal risk categories: **Low Risk, Moderate Risk, and High Risk**.

Parameter	**Value**
Model	MobileNetV2
Input Size	128 × 128 × 3
Classes	3 (Low, Moderate, High)
Training Images	612
Validation Images	108
Test Images	180
Epochs	40
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Transfer Learning	ImageNet Pretrained Weights

Table 1.9 Model Training Configuration

5.1 Dataset Distribution

The dataset was divided into training, validation, and testing subsets.

Dataset Split	**Number of Images**
Training Set	612
Validation Set	108
Test Set	180
Total Images	900

Table 2.0 Dataset Distribution

The dataset contains three classes:

- Low Hormonal Risk
- Moderate Hormonal Risk
- High Hormonal Risk

5.2 MobileNetV2 Architecture Summary

The MobileNetV2 network was used as the backbone feature extractor. The extracted deep facial features were passed through fully connected layers for hormonal risk classification The key goals include:

Layer	Output Shape
Input Layer	128 × 128 × 3
Data Augmentation	128 × 128 × 3
Rescaling Layer	128 × 128 × 3
MobileNetV2 Backbone	4 × 4 × 1280
Global Average Pooling	1280
Dense Layer	64

Table 2.1 Model Training Configuration

Total Parameters

Type	Count
Total Parameters	2,340,163
Trainable Parameters	82,179
Non-Trainable Parameters	2,257,984

Table 2.2 Total Parameters

5.3 Training Performance

The model was trained for 40 epochs. During training, the model gradually improved its classification capability.

Metric	Value
Training Accuracy	81.54%
Validation Accuracy	72.22%
Training Loss	0.4856
Validation Loss	0.5919

Table 2.2 Training Performance

The training and validation curves indicate stable learning behavior with gradual improvement in classification performance.

5.1 Test Performance Evaluation

- The trained MobileNetV2 model was evaluated on an independent test dataset containing 180 facial images.

Metric	Value
Test Accuracy	73.89%
Test Loss	0.5924

Table 2.3 Test Performance Evaluation

5.2 Classification Report

The performance of the model for each hormonal risk category is summarized below

Class	Precision	Recall	F1-Score
High Risk	0.80	0.82	0.81
Low Risk	0.78	0.70	0.74
Moderate Risk	0.65	0.70	0.67

Table 2.4 **Classification Report**

Overall Performance

Metric	Value
Accuracy	0.74
Macro Average Precision	0.74
Macro Average Recall	0.74
Macro Average F1-Score	0.74

Table 2.5 Overall Performance

5.1 Result Analysis

Among all classes, the **High Risk** category achieved the highest performance with an F1-score of **0.81**, indicating that the model successfully learned facial characteristics associated with hormonal imbalance. The **Low Risk** category also showed satisfactory performance with an F1-score of **0.74**. The **Moderate Risk** category achieved comparatively lower performance due to overlapping visual characteristics between neighboring classes.

Overall, the model achieved a test accuracy of **73.89%**, confirming the effectiveness of transfer learning and deep facial feature extraction for hormonal health assessment.

5.2 X XGBoost-Based Hormonal Health Risk Prediction

Environment Setup and Dependency Installation

The XGBoost-based hormonal health prediction models were developed using Python 3.12 within a virtual environment. Required machine learning libraries were installed through the requirements file to ensure a consistent and reproducible execution environment.

The major libraries used include:

- XGBoost 3.2.0
- Scikit-Learn 1.8.0
- Pandas 3.0.2
- NumPy 2.4.4
- OpenPyXL

A dedicated virtual environment was created and activated before installing all dependencies. This isolated environment ensured compatibility among packages and prevented conflicts with existing Python installations.

5.3 Model Training Pipeline

The training pipeline consisted of five independent XGBoost classification models developed for different hormonal health conditions:

1. PCOS Dataset Model
2. PCOD Version-1 Model
3. PCOD Version-2 Model
4. Thyroid Version-2 Model

5. Thyroid Preprocessed Dataset Model

Each model was trained separately using structured clinical and symptom-based datasets.

The training process included:

- Dataset loading
- Data preprocessing
- Feature selection
- Train-test split
- XGBoost model training
- Performance evaluation
- Feature importance analysis
- Model serialization and storage

After successful training, each model was saved in JSON format for future deployment

5.4 PCOS Dataset Model

Dataset Information

The PCOS dataset contained 32 clinical and symptom-based features related to hormonal health

Parameter	**Value**
Training Samples	540
Testing Samples	136
Features	32

Table 2.6 Dataset Information

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Performance Metrics

The XGBoost classifier achieved strong performance in identifying PCOS-related hormonal conditions.

Metric	**Value**
Accuracy	93.38%
AUC Score	97.66%

Table 2.7 Performance Metrics

Classification Report

Class	**Precision**	**Recall**	**F1-Score**
Non-PCOS	0.83	0.91	0.87
PCOS	0.97	0.94	0.96

Table 2.8 Classification Report

The model demonstrated excellent capability in distinguishing between PCOS and non-PCOS cases with minimal classification errors.

Important Features

The most influential features identified by XGBoost were:

Feature	Importance
Cyst Ovary	0.1798
Period Type	0.1745
Hormonal Imbalance	0.1364
Dark Area	0.0365
Excess Facial Hair	0.0355
Family Background	0.0303
Excess Body Hair	0.0273
Losing Hair	0.0236
LH Hormone	0.0204

Table 2.9 Model Training Configuration

These features indicate that ovarian cyst presence, menstrual irregularities, and hormonal imbalance are major indicators of PCOS.

Model Storage

The trained model was saved as:

pcos_dataset_xgb.json

5.5 PCOD Version-2 Model Results

Dataset Information

Parameter	Value
Training Samples	3200
Testing Samples	800
Features	10

Table 3.0 Dataset Information

Performance Metrics

Metric	Value
Accuracy	94.63%
AUC Score	97.40%

Table 3.1 Performance Metrics

Classification Report

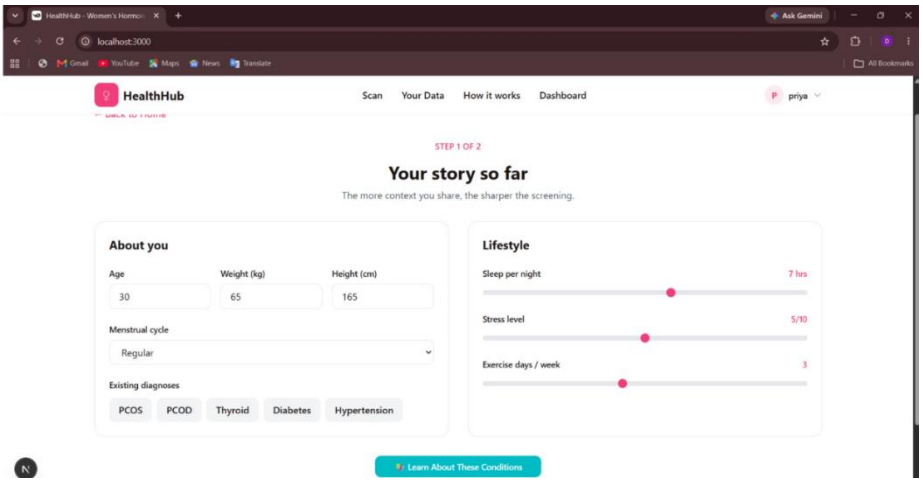
Class	Precision	Recall	F1-Score
Negative	0.96	0.98	0.97
Positive	0.78	0.67	0.72

Table 3.2 Classification Report

The model successfully classified PCOD cases with high overall accuracy while maintaining robust discrimination capability

Feature	Importance
Irregular Periods	0.5508
Weight Gain	0.1192
Cycle Length Days	0.0700
Acne	0.0562
Difficulty Getting Pregnant	0.0509
Excess Facial/Body Hair	0.0498
Hair Loss	0.0318
BMI	0.0310
BMI Category	0.0224
Age	0.0180

Table 3.3 Important Features



Model Storage

The trained model was saved as:

pcod_v2_xgb.json

5.5.1 PCOD Version-1 Model Results

Dataset Information

Parameter	Value
Training Samples	2400
Testing Samples	600
Features	9

Table 3.4 Model Training Configuration

Performance Metrics

Metric	Value
Accuracy	93.00%
AUC Score	97.80%

Table 3.5 Model Training Configuration

Classification Report

Class	Precision	Recall
Negative	0.94	0.94
Positive	0.90	0.90

Table 3.6 Model Training Configuration

The model demonstrated balanced performance across both classes with minimal bias.

Important Features

Feature	Importance
Irregular Periods	0.5847
Excess Facial/Body Hair	0.1086
Weight Gain	0.0772
Cycle Length Days	0.0544
Difficulty Getting Pregnant	0.0509
Acne	0.0436
Hair Loss	0.0420
BMI	0.0220
Age	0.0165

Again, irregular menstrual patterns emerged as the strongest indicator

5.5.2 Thyroid Version-2 Model Results

Dataset Information

The thyroid dataset consisted of three classes:

- Hyperthyroid
- Hypothyroid
- Normal

Parameter	Value
Training Samples	3200
Testing Samples	800
Features	17

Table 3.7 Dataset Information

Performance Metrics

Metric	Value
Accuracy	99.25%
AUC Score	99.96%

Table 3.8 Performance Metrics

Classification Report

Class	Precision	Recall	F1-Score
Hyperthyroid	1.00	1.00	1.00
Hypothyroid	0.99	0.97	0.98
Normal	0.99	1.00	0.99

Table 3.9 Classification Report

The model achieved near-perfect classification performance across all thyroid conditions.

5.5.3 Thyroid Preprocessed Dataset Results

Dataset Information

Parameter	Value
Training Samples	7337
Testing Samples	1835
Features	40

Table 4.0 Dataset Information

Performance Metrics

Metric	Value
Accuracy	95.31%
AUC Score	99.40%

Table 4.1 Performance Metrics

Performance Metrics

Metric	Value
Accuracy	95.31%
AUC Score	99.40%

Table 4.2 Performance Metrics

Classification Report

Class	Precision	Recall	F1-Score
Hyperthyroid	0.86	0.79	0.83
Hypothyroid	0.99	0.98	0.99
Negative	0.96	0.98	0.97
Other	0.89	0.85	0.87

Table 4.3 Classification Report

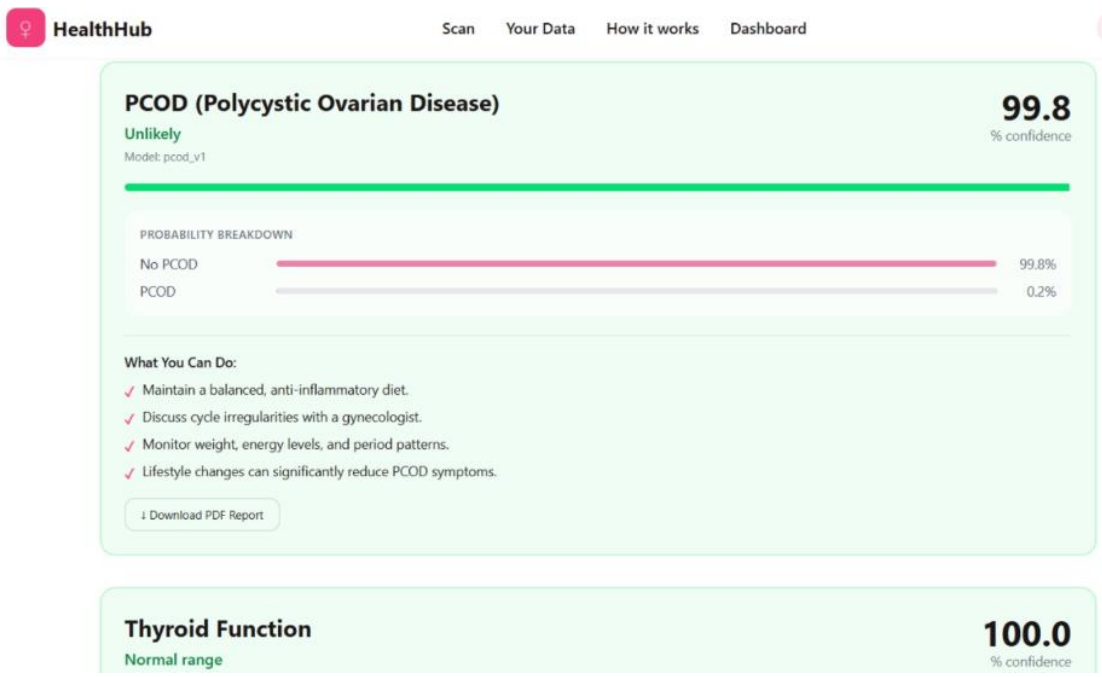
The model demonstrated excellent generalization capability across multiple thyroid-related conditions.

Summary of XGBoost Results

Model	Accuracy	Model
PCOS Dataset	93.38%	PCOS Dataset
PCOD V1	93.00%	PCOD V1
PCOD V2	94.63%	PCOD V2
Thyroid V2	99.25%	Thyroid V2
Thyroid Preprocessed	95.31%	Thyroid Preprocessed

Table 4.4 Summary of XGBoost Results

Among all structured-data models, the Thyroid V2 XGBoost classifier achieved the highest performance with an accuracy of **99.25%** and an AUC score of **99.96%**. The experimental results demonstrate the effectiveness of XGBoost in learning complex relationships between clinical symptoms, hormonal indicators, and diagnostic outcomes. These models form an important component of the proposed AI-assisted hormonal health assessment system.





Women's Health Prediction Report

Generated June 23, 2026 at 09:00 AM

Patient Information

Name	priya
Email	deekshadeeksha3396@gmail.com
Phone	+917019492312

Prediction Result

Condition	THYROID
Model used	thyroid_v2
Result	normal
Confidence	100.0%
Date & time	June 23, 2026 at 08:59 AM

Full Probability Breakdown

Class	Probability
hyperthyroid	0.0%
hypothyroid	0.0%
normal	100.0%

Recommendations

- Discuss a full thyroid panel (TSH, T3, T4) with your doctor if you haven't had one recently.
- Note any changes in weight, energy, heart rate, or temperature sensitivity.
- Thyroid conditions are very manageable with proper medical care once diagnosed.

This report is generated by an automated prediction system and is for informational purposes only. It is not a medical diagnosis. Please consult a qualified healthcare professional for proper evaluation, diagnosis, and treatment.

CONCLUSIONS

This project developed an AI-assisted hormonal health risk prediction system using facial images and textual health data. MobileNetV2 was used to extract facial features, while XGBoost was employed for the classification of hormonal health conditions such as PCOS, PCOD, and thyroid disorders.

The experimental results demonstrated high prediction accuracy, indicating the effectiveness of combining deep learning and machine learning techniques. The system successfully classified hormonal risk levels and identified important health-related features from both image and text data.

Overall, the proposed framework provides a non-invasive and efficient approach for early hormonal health assessment, supporting timely medical consultation and improving awareness of hormonal disorders.

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